

Mahmoud Bassem Badr

Software Engineer

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Education

Ain Shams University, Cairo - BS in Computer Sciences

October 2022 - June 2026

- Relevant Coursework: Algorithms, Data Structures, Operating Systems, Object-Oriented Programming, Computer Architecture, Mathematics, Compilers Theory, Machine Learning.

Experience

Competitive Programmer for NEAR

September 2024 – November 2024

- Worked on problem-solving tasks to create an AI model in C++.
- Created tutorials for solving algorithms, summarized statements of tasks, and added new test cases.

Projects

FOS - A Simple Operating System - [LINK](#)

- Engineered kernel and user heap allocators via a first-fit strategy, accelerating allocation by 85% and reducing memory by 25%.
- Built a shared object memory system with robust mutex and locking mechanisms, improving data integrity by 20%.
- Technologies: C, x86 Assembly, Low-Level Memory Management, Concurrency Control.

Image Encryption/Compression - [LINK](#)

- Encrypted images using LFSR, securing 500+ daily uploads against transmission threats.
- Implemented Huffman-based image compression, shrinking storage by 200%.
- Technologies: C#, WinForms.

Guide Me - [LINK](#)

- Integrated graph update mechanics and optimized path-finding using advanced algorithms, boosting performance 5x.
- Technologies: C++, SFML, CMake.

Desktop Flight Booking - [LINK](#)

- Coordinated with 5 peers to deliver a robust desktop booking system under a 3-week deadline.
- Developed Seat selection and history features, streamlining user flow by 30%.
- Accelerated database lookups using Trie-based indexing, cutting user search time by 300%.
- Technologies: Object-Oriented Programming (OOP), Java, JavaFX, CSS.

Social Media Sentiment Analysis - [LINK](#)

- Constructed a custom Support Vector Machine model, achieving 76.88% accuracy on sentiment classification.
- Technologies: Python, Tkinter.

Awards

ICPC ID

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|---|----------------------|
| • 38th Place in ICPC Regionals (Africa and Arab Collegiate Programming Contest) | <i>December 2025</i> |
| • ICPC Nationals Bronze Medalist (Egyptian Collegiate Programming Contest) and 16th Place | <i>August 2025</i> |
| • 6th Place in ICPC Nationals (Egyptian Collegiate Programming Contest) Qualification Day 5 | <i>August 2025</i> |

Skills

Programming Languages: C, C++, Java, C#, Python, SQL.

Technical Concepts: OOP, Data Structures, Algorithms, Operating Systems, Relational Databases, Networks.

Developer Tools: Git, GitHub, Linux.

Languages: *Native:* Arabic, *proficient:* English, *Elementary:* German

Extracurricular Activities

acmASCIS, Cairo

September 2019 – Present

- Worked as student mentor guiding freshmen in college to participate in programming contests and teaching them algorithms and data structures.